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Structural Causal Models

A Shared Language for Scientific Inquiry

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https://julianfrattini.github.io/

JulianFrattini/**bda4sci** Public

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Directed Acyclic Graphs

A Syntax to Formalize Causal Assumptions

Causal Modelling

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A (structural) causal model is a **conceptual model** that *represents the causal mechanisms of a system* [1]. They serve two main purposes [2]:

1. **Visualizing causal assumptions** and determining testable implications.
2. **Proving the possibility** of computing causal effects from observational data.

Structure of this Seminar

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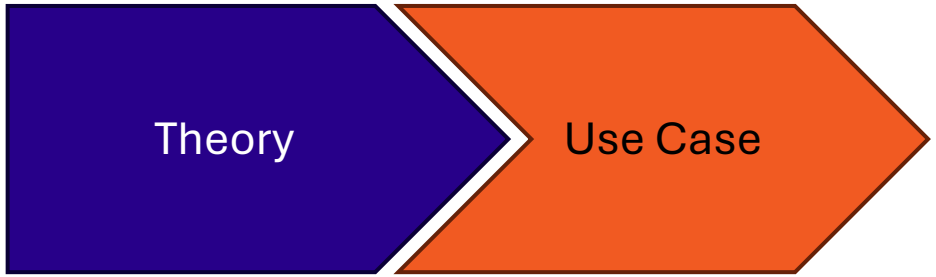
Every lesson consists of its theory and one or more use cases.



Directed Acyclic Graphs

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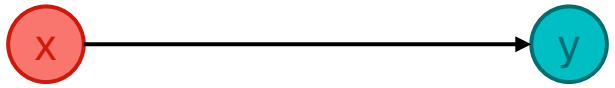
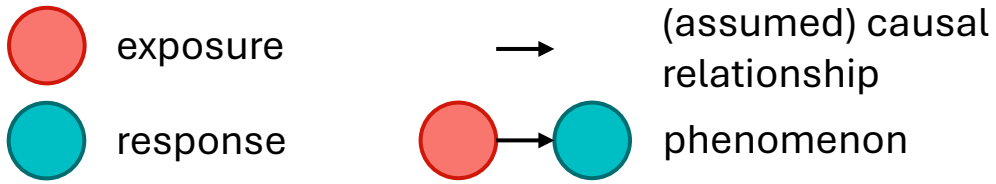
Causal models can be realized as directed, acyclic graphs (DAGs).





Theory Use Case

Formulating Casual Hypotheses

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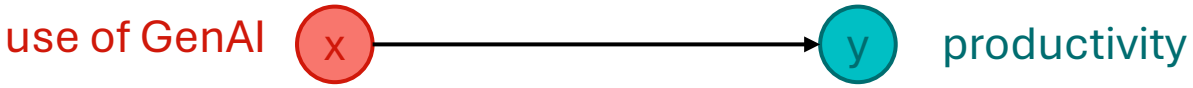
DAGs convey complex relationships between multiple variables in a simple and straightforward manner.

> RQ: Does the use of **generative AI (GenAI)** have **an effect** on the **productivity** of developers?



Directed Acyclic Graphs

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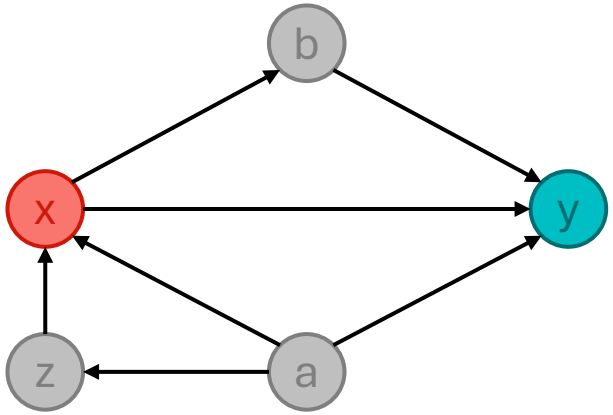

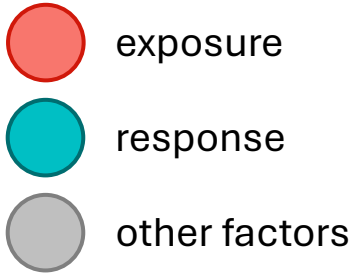

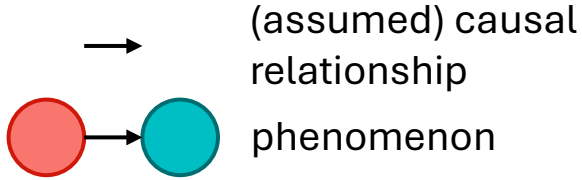

Theory Use Case

Formulating Complex Hypotheses

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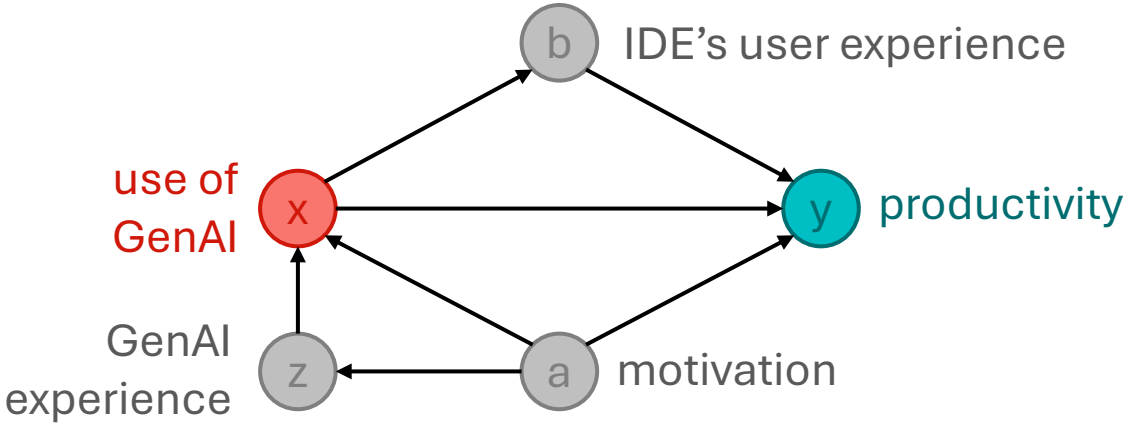
DAGs convey complex relationships between multiple variables in a simple and straightforward manner.

> RQ: Does the use of **generative AI (GenAI)** have **an effect** on the **productivity** of developers, assuming that **motivation** affects both usage (moderated by **GenAI experience**) and productivity, and accounting for an indirect effect of the **IDE's user experience**?



Disputing Causal Hypotheses

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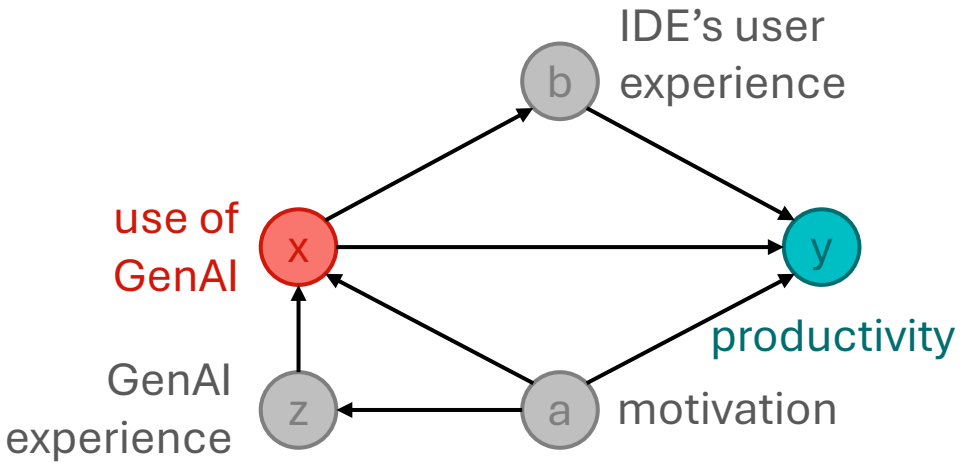
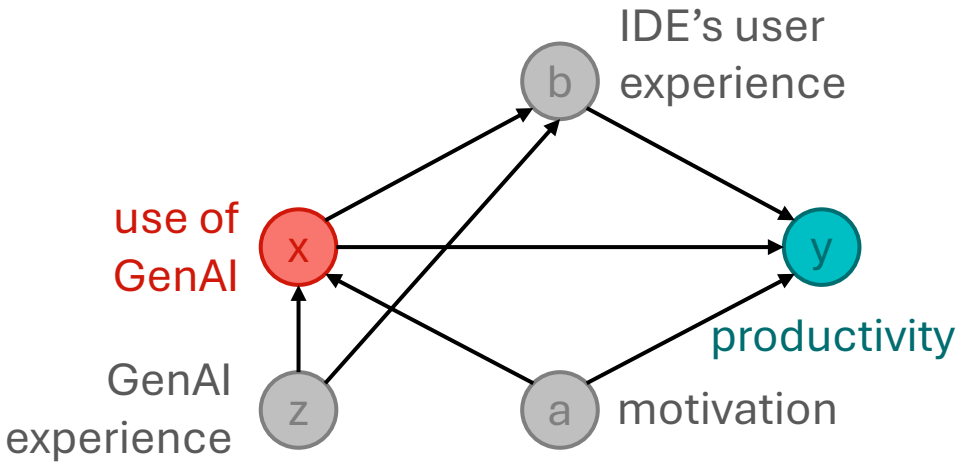
The explicit formulation of causal hypotheses in DAGs makes it easy to compare and contrast assumptions.





These comparisons can inform the design of an empirical study to verify, which of the two hypotheses reflects reality better.

Distinguishing Study Types

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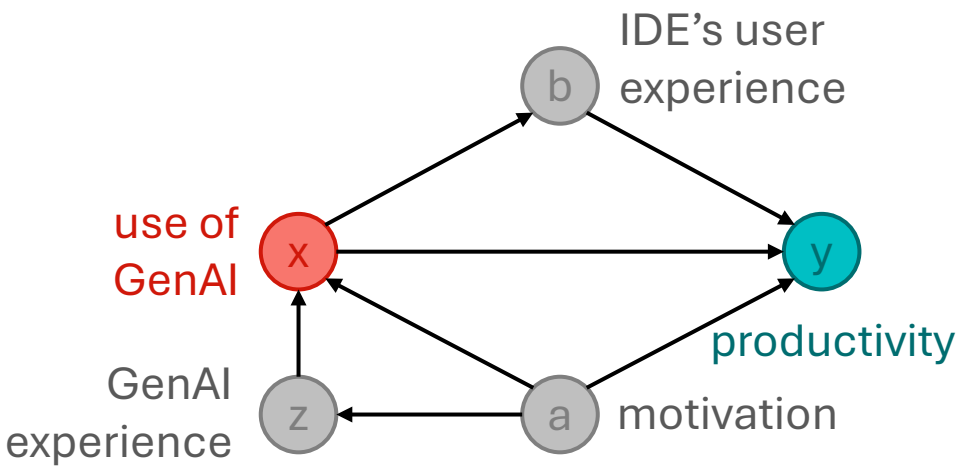
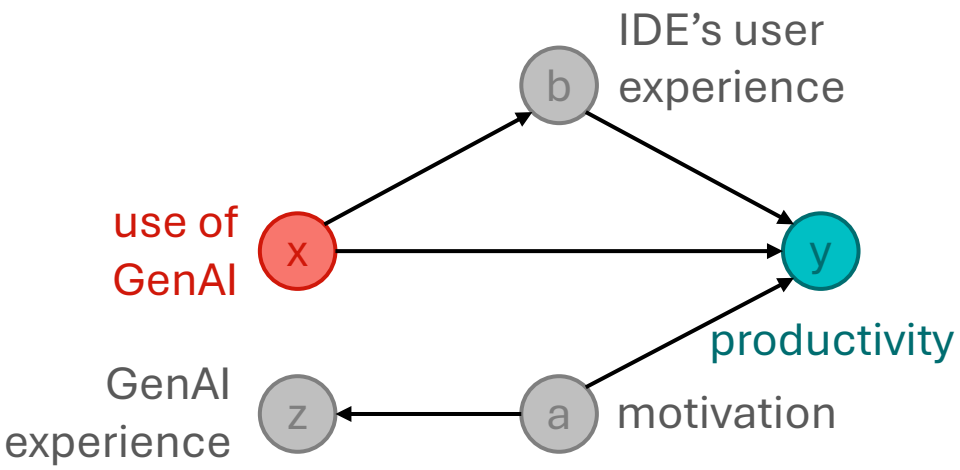
The causal DAG also reveals controlled vs. observational studies: a controlled study is indicated by a DAG where no variable affects the exposure.

Observational study



Controlled Experiment



Types of Relationships

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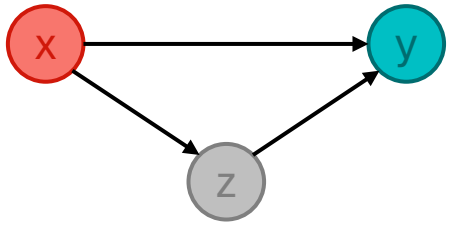
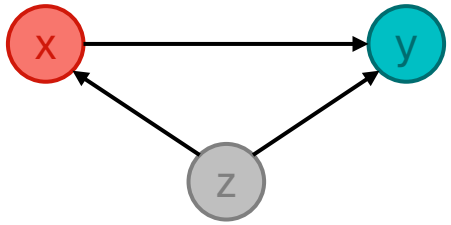
When a DAG involves more than two variables, certain relationships can emerge:

Mediation



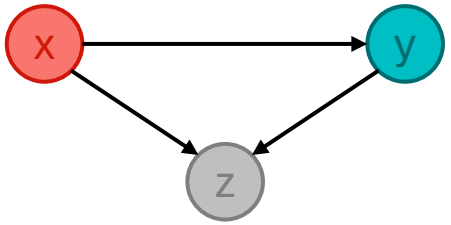
z *mediates* the effect of x on y. There is a direct (x → y) and an indirect (x → z → y) effect.

Confounding



z *confounds* the effect of x on y. An observed effect (x → y) might be caused by the confounder (x ← z → y).

Collider



z is a common effect of x and y.

Dealing with Variables

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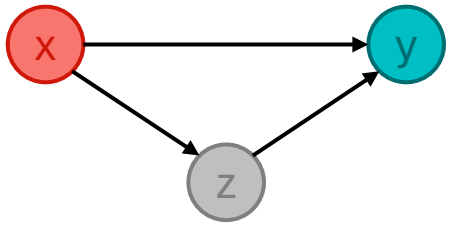
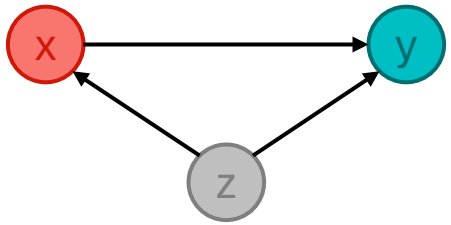
Based on the type of relationship between variables, these additional factors need to be treated differently.

Mediation



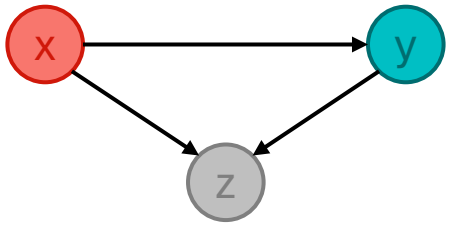
You **can** control z if you want to distinguish the direct from the indirect effect

Confounding



You **must** control z to avoid confounding

Collider



You **must not** control z to avoid confounding

Control can either be by design (fixing variables) or by analysis (stratifying the data).

Encoding Observability

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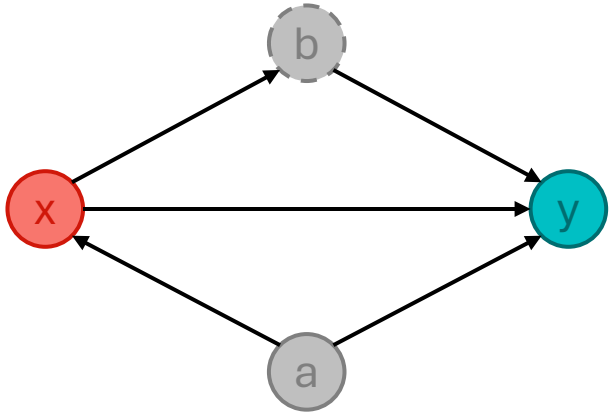
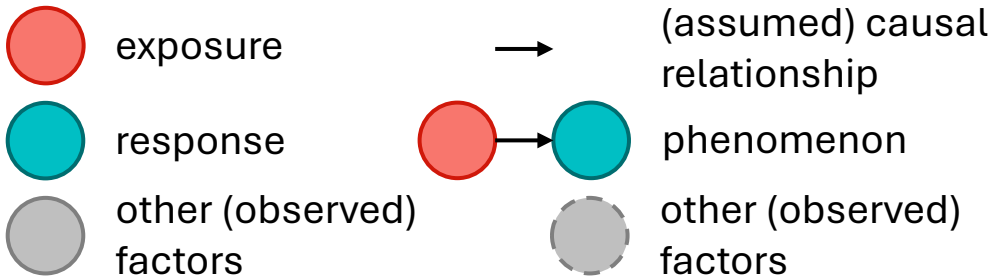
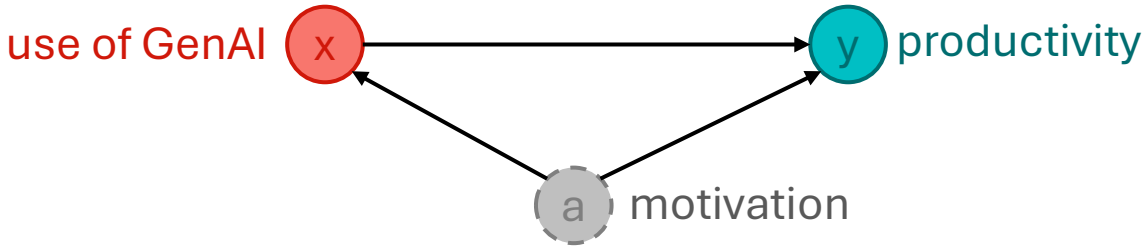
Nodes in DAGs can indicate whether a variable is observable or not.





Determining Study-ability

Causal DAGs allow deciding whether it is even possible to determine the (unbiased) causal effect of a phenomenon.



If a phenomenon has an unobserved confounder, then the causal effect of the exposure on the outcome cannot be estimated from observational data.

Visualizing Moderation

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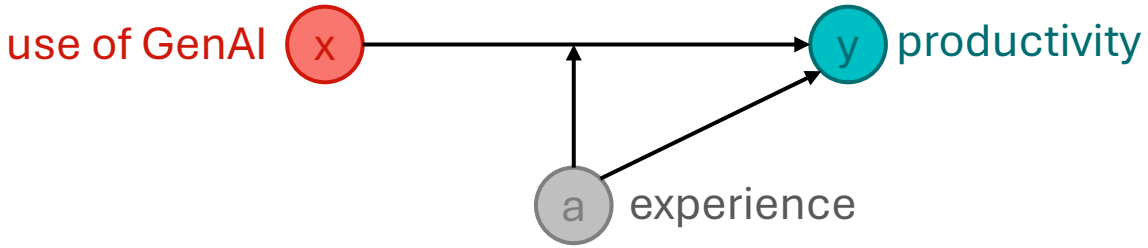
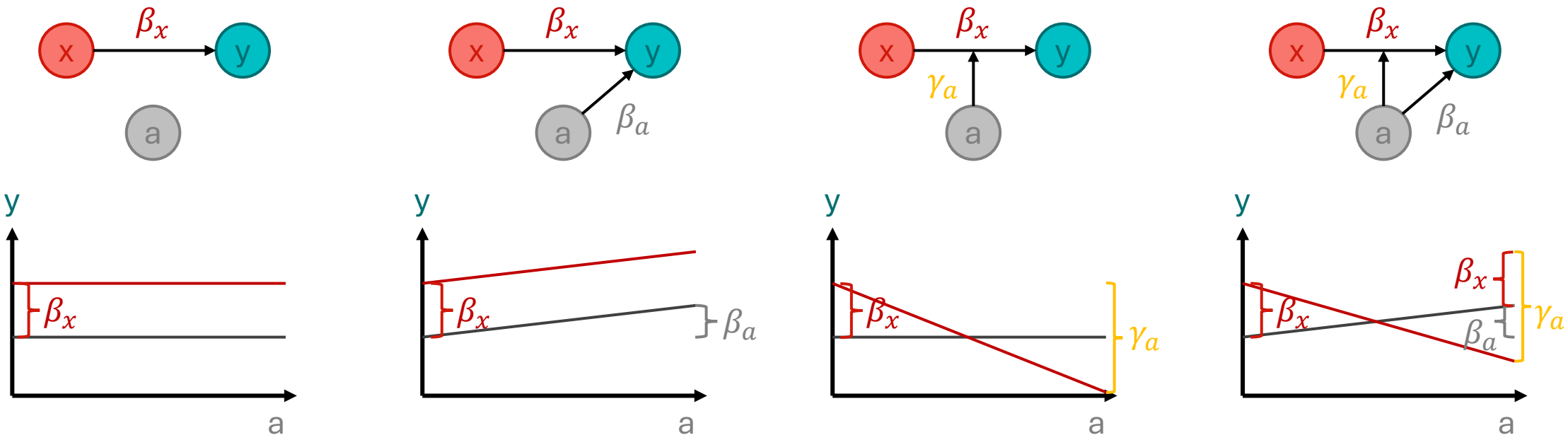
Factors can not only affect other factors, but also other relationships in a process called moderation.



A moderator changes the effect that one variable has on another.

Determining Moderation

The existence of a moderator evokes a distinct effect pattern.



Exercises

Applying Causal Modelling

Exercise: Formulating a DAG

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Derive a causal DAG from the following description:

“Researchers are interested in knowing if using ChatGPT affects the quality and length of code reviews at a company. Since they cannot interrupt the regular workflow, they cannot force anyone to use or not use ChatGPT. They do know, however, that younger developers use ChatGPT more often. On the other hand, they know that more skilled developers generally produce more concise (i.e., shorter) and better code reviews.”

Solution: Formulating a DAG

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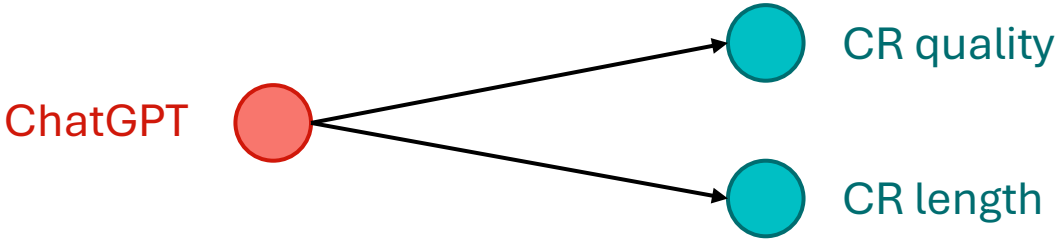
"Researchers are interested in knowing if **using ChatGPT** affects the **quality** and **length of code reviews** at a company. Since they cannot interrupt the regular workflow, they cannot force anyone to use or not use ChatGPT. They do know, however, that younger developers use ChatGPT more often. On the other hand, they know that more skilled developers generally produce more concise (i.e., shorter) and better code reviews."



Solution: Formulating a DAG

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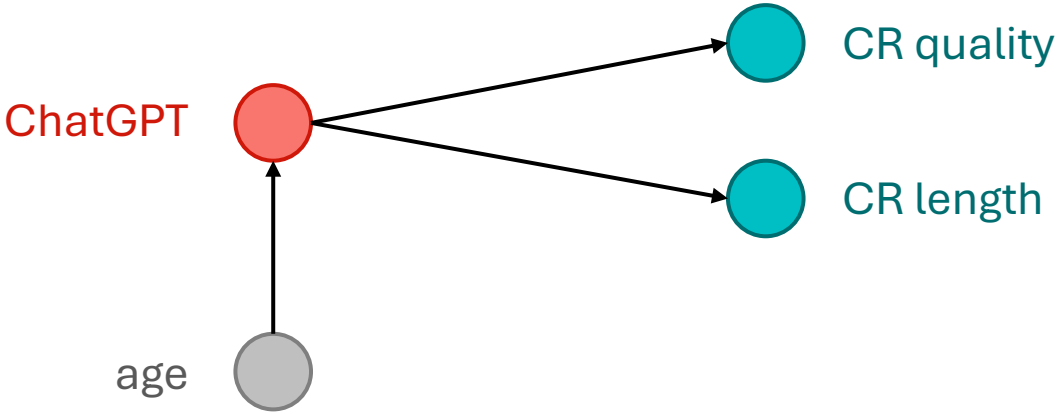

Solution: Formulating a DAG

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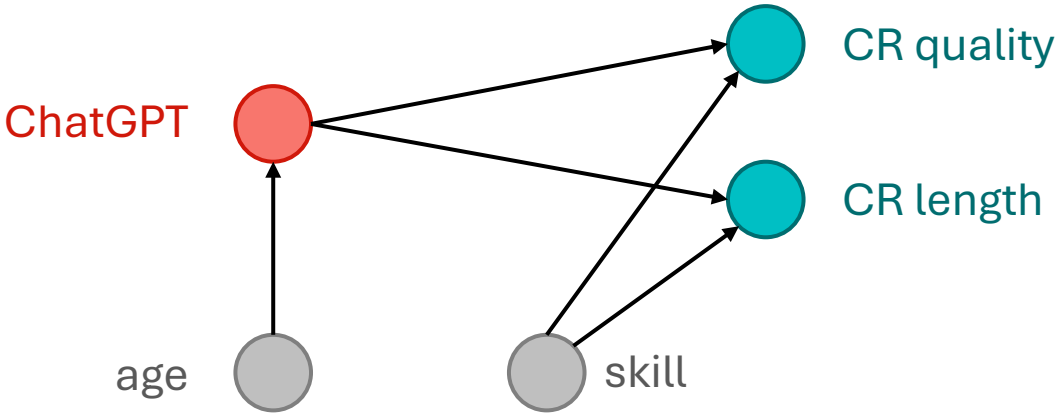

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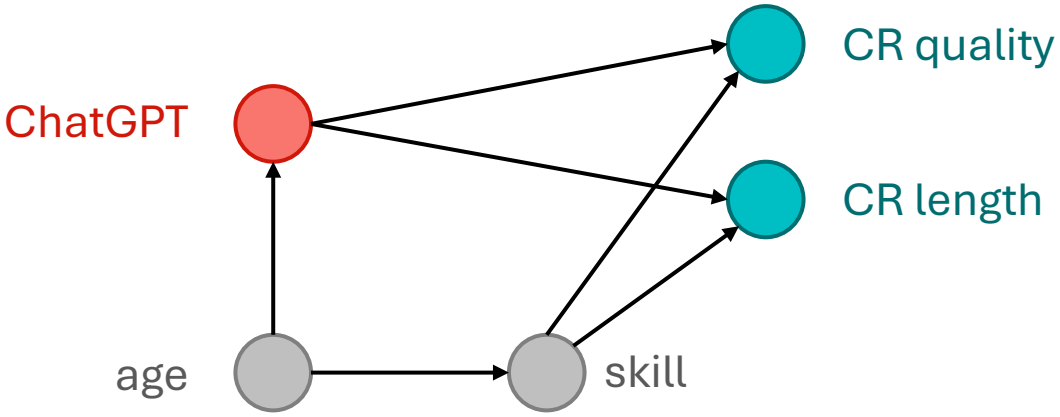

Material

JulianFrattini/**bda4sci** Public

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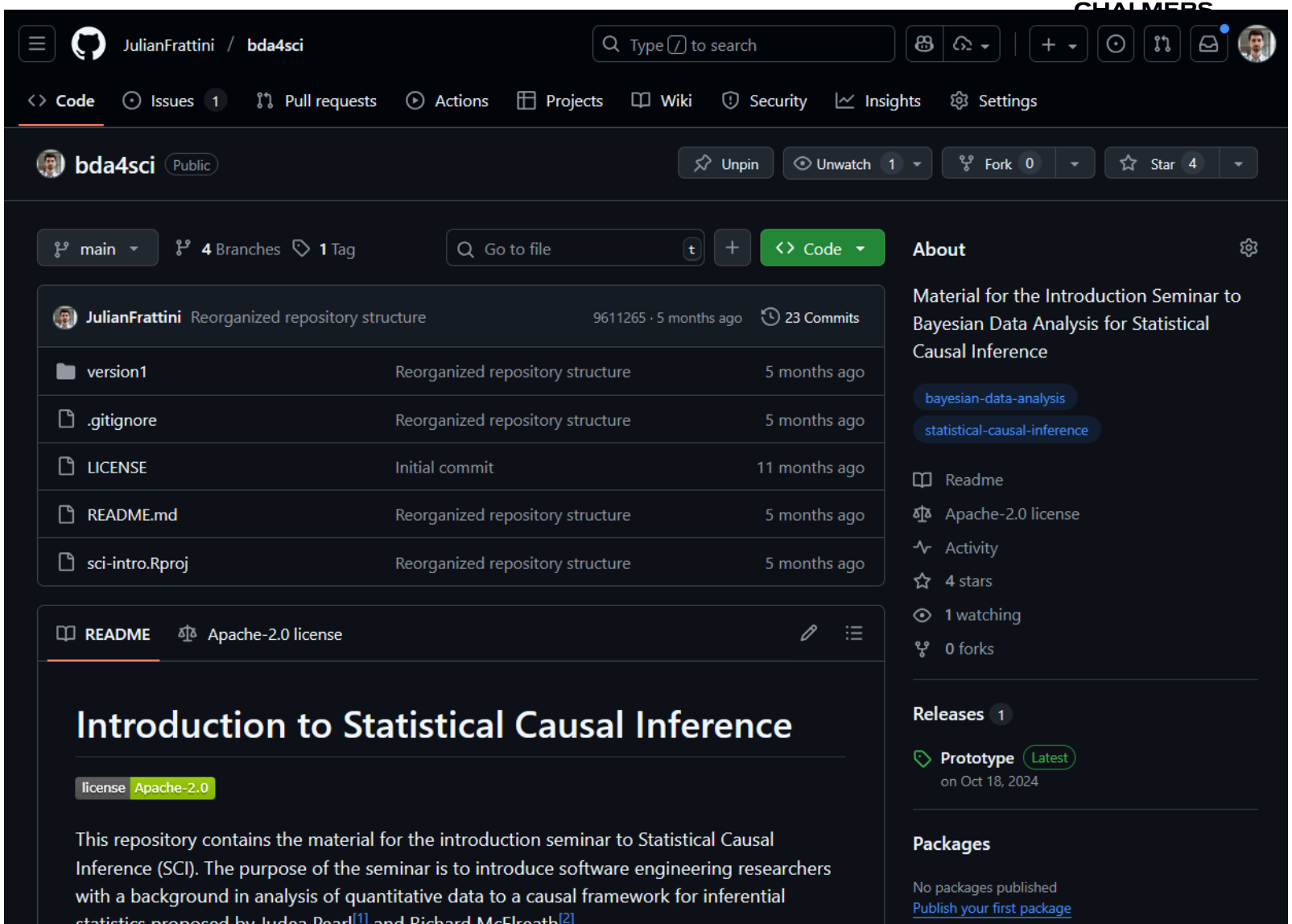
All material is public and open-source on GitHub. Feel free to clone, fork, and share the study material.

Comments, requests, and additions are welcome.



References

[1] Pearl, J. (2009). *Causality*. Cambridge university press.

[2] Cinelli, C., Forney, A., & Pearl, J. (2024). A crash course in good and bad controls. *Sociological Methods & Research*, *53*(3), 1071-1104.